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Course: CSE LAB

Course Code: 1202

Assignment Topic:

1. Showing “Hello, world” in Output using C language
2. Summation of any 2 numbers using C language
3. Summation, Subtraction, Multiplication, Division of any 2 numbers using C language

Submitted to:

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1. Showing Hello World in output in using C language:

- **Program:**

```
//first program in c
#include <stdio.h>
//function main begins program execution
int main(void)
{
  printf("Hello, World!!");    //print the statement
   return 0;  //indicate that program ended successfully
}     //end function main
```

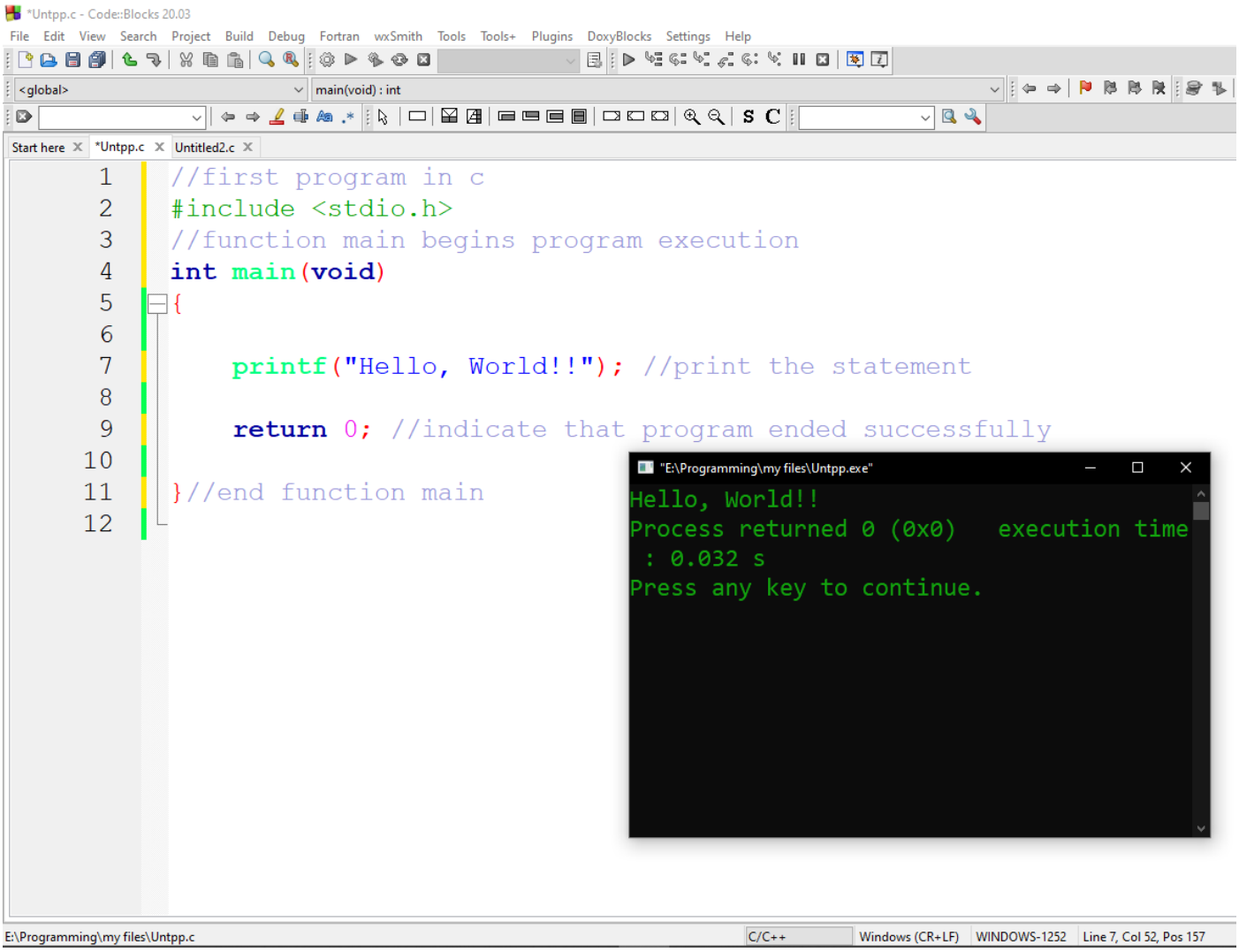

Figure 1.1 Program

- **Output:**

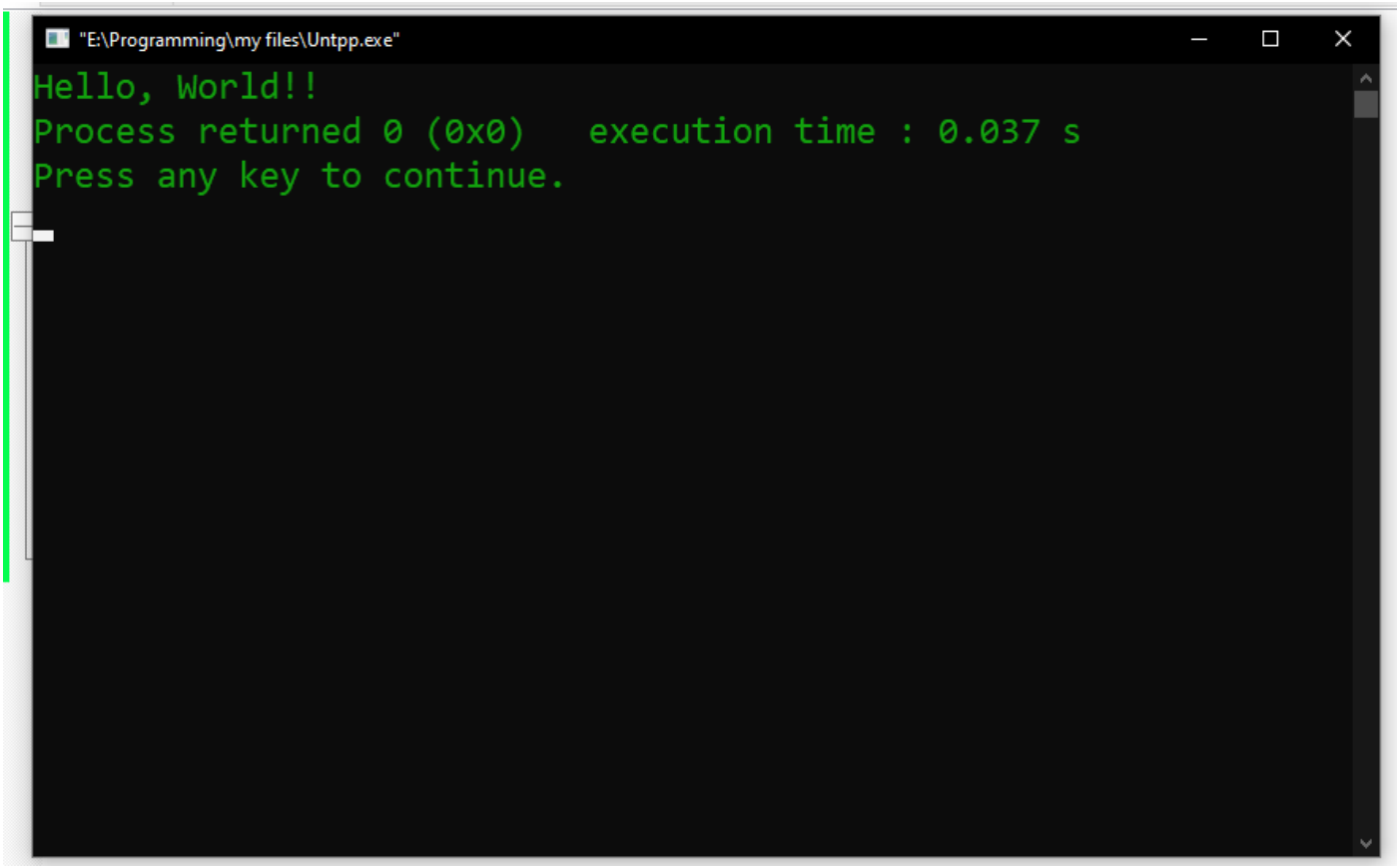

Figure 1.2 Output

- **Description:**

I begin by considering a simple C program in *Figure 1.1*

Begin with **//** indicating that these lines a comment, **"first program in C".** I insert comments to document programs and improve program readability. My Comments do not cause the computer to perform any action when the program is run.

In the 1st line I used a comment for indicate the purpose of this code. In line 2 I used a library function called **#include <stdio.h>** which is a directive to the C preprocessor**.** In the 4th line the main function begins execute the program. In the main function **int main (void)** all the function were written enter in the curly brace {}.

In line 7 I use **printf** for showing "Hello, world" in the monitor. It indicates that **printf** is supposed to do something out of the ordinary. The output is also here, as we can see the output in *Figure 1.2*

In line 9 I indicate that the program ended successfully by using **return 0;**

2. Summation of any 2 numbers using C language:

- **Program:**

```
//Summation of 2 numbers
#include <stdio.h>
//function main begins program execution
int main()
{
   int x, y, sum;                                  //assigning variables
   printf("Enter your first number:\n");          //user prompt
   scanf("%d", &x);                               //taking input from user
   printf("Enter your 2nd number:\n");            //user prompt
   scanf("%d", &y);                               //taking input from user
   sum=x+y; //summation of 2 numbers
   printf("The summation is:%d\n",sum);           //showing output
    return 0;                                     //indicate that program ended successfully
}                                                 //end function main
```

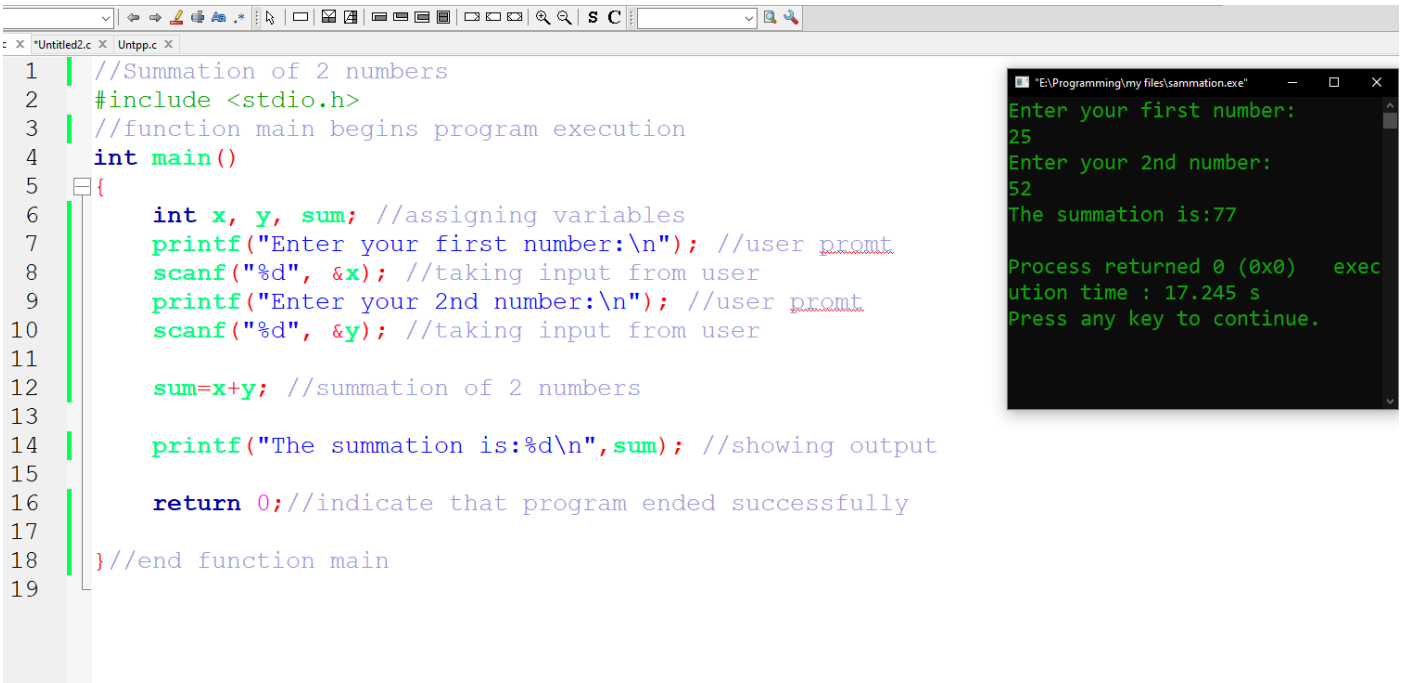

Figure 2.1 Program

- **Output:**

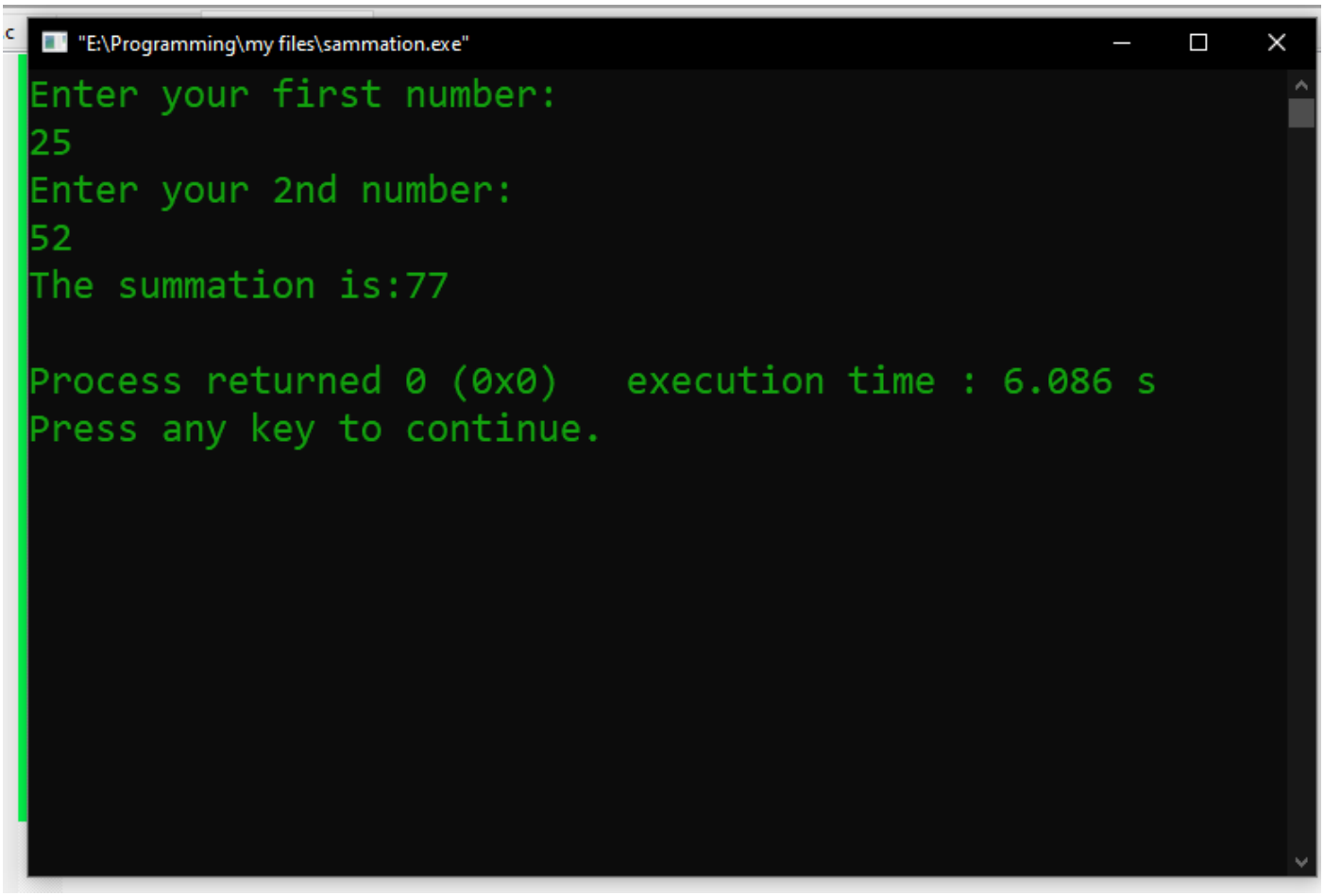

Figure 2.2 Output

- **Description:**

Begin with **//** indicating that this one line is a comment, **"Summation of 2 numbers".** I insert comments to document programs and improve program readability. My Comments do not cause the computer to perform any action when the program is run.

In the 1st line I used a comment for indicate the purpose of this code. In line 2 I used a library function called **#include <stdio.h>** which is a directive to the C preprocessor**.** In the 4th line the main function begins execute the program. In the main function **int main ()** all the function were written enter in the curly brace {}.

The names **x**, **y** and **sum** are the names of variables**.** In this program first of all I used **int x, y, sum;** This means that it needs to be worked on those variables. Then I used **printf** for user prompt, **"Enter the first number"**. Our next program uses the Standard Library function **scanf** to obtain two integers typed by a user at the keyboard, computes the sum of these values and prints the result using **printf** for showing "**The summation**" in the monitor. The output is also here, as we can see the output in *Figure 2.2*

In line 16 I indicate that the program ended successfully by using **return 0;**

3. Summation, Subtraction, Multiplication, Division of any 2 numbers using C language

- **Program:**

```
//summation, subtraction, Multiplication, Division of 2 numbers
#include <stdio.h>
//function main begins program execution
int main()
{
   int x, y, sum, sub, multi, div;                    //assigning variable
   printf("Enter your first number:\n");              //user promt
   scanf("%d", &x);//taking input from user
   printf("Enter your 2nd number:\n");            //user promt
   scanf("%d", &y);//taking input from user
   sum=x+y; //summation of 2 numbers
   sub=x-y; //subtraction of 2 numbers
   multi=x*y; //multiplication of 2 numbers
   div=x/y; //division of 2 numbers
   printf("The summation is:%d\n",sum);          //showing output
   printf("The subtraction is:%d\n",sub);
   printf("The multiplication is:%d\n",multi);
   printf("The division is:%d",div);
return 0; //indicate that program ended successfully
}//end function main
```

```c
//summation, subtraction, Multiplication, Division
#include <stdio.h>
//function main begins program execution
int main()
{
    int x, y, sum, sub, multi, div; //assigning variables
    printf("Enter your first number:\n");//user promt
    scanf("%d", &x);//taking input from user
    printf("Enter your 2nd number:\n");//user promt
    scanf("%d", &y);//taking input from user
    sum=x+y; //summation of 2 numbers
    sub=x-y; //subtraction of 2 numbers
    multi=x*y; //multiplication of 2 numbers
    div=x/y; //division of 2 numbers
    printf("The summation is:%d\n",sum); //showing output
    printf("The subtraction is:%d\n",sub);
    printf("The multiplication is:%d\n",multi);
    printf("The division is:%d",div);
return 0; //indicate that program ended successfully

}//end function main
```

Figure 3.1 Program

- **Output:**

```
"E:\Programming\my files\Untitled2.exe"
Enter your first number:
12
Enter your 2nd number:
58
The summation is:70
The subtraction is:-46
The multiplication is:696
The division is:0
Process returned 0 (0x0)   execution time : 14.583 s
Press any key to continue.
```

Figure 3.2 Output

- **<u>Description:</u>**

Begin with **//** indicating that this one line is a comment, **"summation, subtraction, Multiplication, Division of 2 numbers".** I insert comments to document programs and improve program readability. My Comments do not cause the computer to perform any action when the program is run.

In the 1st line I used a comment for indicate the purpose of this code. In line 2 I used a library function called **#include <stdio.h>** which is a directive to the C preprocessor**.** In the 4th line the main function begins execute the program. In the main function **int main ()** all the function were written enter in the curly brace {}.

The names **x**, **y**, **sum, sub, multi, div** are the names of variables**.** In this program first of all I used **int x, y, sum, sub, multi, div;** This means that, it needs to be worked on those variables. Then I used **printf** for user prompt, **"Enter the first & second number"**. Our next program uses the Standard Library function **scanf** to obtain two integers typed by a user at the keyboard, computes the sum of these values and prints the result using **printf** for showing **The summation, subtraction, multiplication & division** one by one in the monitor. In the line 11 to 14 those Summation, subtraction, multiplication, division functions are written for processes the integer values of x & y variables. After processing the value, the output is also here, as we can see the output in *Figure 3.2*

In line 19 I indicate that the program ended successfully by using **return 0;**

THE END